

Host Discovery options using Nmap and hping3

Here I am choosing a Vulnerable Machine available from Tryhackme as the target.

This is the Target : **10.10.43.70**

So let's start by Pinging the target

1. \$ nmap -sn 10.10.43.70

```
(kali㉿kali) - [~/Downloads]
$ sudo nmap -sn 10.10.43.70
[sudo] password for kali:
Starting Nmap 7.94SVN ( https://nmap.org ) at 2025-03-24 06:40 EDT
Nmap scan report for 10.10.43.70
Host is up (0.16s latency).
Nmap done: 1 IP address (1 host up) scanned in 11.30 seconds
```

It performs a ping sweep - checks which hosts are alive without port scanning.

2. \$ nmap -PE 10.10.43.70

```
(kali㉿kali) - [~/Downloads]
$ sudo nmap -PE 10.10.43.70
Starting Nmap 7.94SVN ( https://nmap.org ) at 2025-03-24 06:40 EDT
Nmap scan report for 10.10.43.70
Host is up (0.15s latency).
Not shown: 991 closed tcp ports (reset)
PORT      STATE SERVICE
135/tcp    open  msrpc
139/tcp    open  netbios-ssn
445/tcp    open  microsoft-ds
3389/tcp   open  ms-wbt-server
49152/tcp  open  unknown
49153/tcp  open  unknown
49154/tcp  open  unknown
49158/tcp  open  unknown
49160/tcp  open  unknown

Nmap done: 1 IP address (1 host up) scanned in 17.16 seconds
```

This sends ICMP request.

3. \$ nmap -PS 80 10.10.43.70

```
(kali㉿kali) - [~/Downloads]
$ nmap -PS 80 10.10.43.70
Starting Nmap 7.94SVN ( https://nmap.org ) at 2025-03-24 06:47 EDT
Nmap scan report for 10.10.43.70
Host is up (0.16s latency).
Not shown: 991 closed tcp ports (conn-refused)
PORT      STATE SERVICE
135/tcp    open  msrpc
139/tcp    open  netbios-ssn
445/tcp    open  microsoft-ds
3389/tcp   open  ms-wbt-server
49152/tcp  open  unknown
49153/tcp  open  unknown
49154/tcp  open  unknown
49158/tcp  open  unknown
49160/tcp  open  unknown

Nmap done: 2 IP addresses (1 host up) scanned in 30.25 seconds
```

Sends a TCP SYN packet to the specified port(s) to check if the host is up.

4. \$ nmap -PA 80 10.10.43.70

This sends the TCP ACK Packets - which is useful for bypassing firewall.

5. \$ nmap -PR 10.10.43.70

This sends ARP requests to discover hosts, using ARP as a medium to scan the network.

hping3

1. hping3 -1 10.10.43.70

```
(kali㉿kali) - [~/Downloads]
$ sudo hping3 -1 10.10.43.70
HPING 10.10.43.70 (tun0 10.10.43.70): icmp mode set, 28 headers + 0 data bytes
len=28 ip=10.10.43.70 ttl=124 id=6024 icmp_seq=0 rtt=164.7 ms
len=28 ip=10.10.43.70 ttl=124 id=6025 icmp_seq=1 rtt=198.5 ms
len=28 ip=10.10.43.70 ttl=124 id=6026 icmp_seq=2 rtt=164.0 ms
len=28 ip=10.10.43.70 ttl=124 id=6027 icmp_seq=3 rtt=169.7 ms
len=28 ip=10.10.43.70 ttl=124 id=6028 icmp_seq=4 rtt=199.9 ms
len=28 ip=10.10.43.70 ttl=124 id=6029 icmp_seq=5 rtt=158.0 ms
len=28 ip=10.10.43.70 ttl=124 id=6030 icmp_seq=6 rtt=172.2 ms
len=28 ip=10.10.43.70 ttl=124 id=6031 icmp_seq=7 rtt=182.4 ms
len=28 ip=10.10.43.70 ttl=124 id=6032 icmp_seq=8 rtt=284.0 ms
len=28 ip=10.10.43.70 ttl=124 id=6033 icmp_seq=9 rtt=181.1 ms
len=28 ip=10.10.43.70 ttl=124 id=6034 icmp_seq=10 rtt=213.5 ms
len=28 ip=10.10.43.70 ttl=124 id=6035 icmp_seq=11 rtt=165.1 ms
len=28 ip=10.10.43.70 ttl=124 id=6036 icmp_seq=12 rtt=211.4 ms
len=28 ip=10.10.43.70 ttl=124 id=6037 icmp_seq=13 rtt=170.4 ms
len=28 ip=10.10.43.70 ttl=124 id=6038 icmp_seq=14 rtt=209.4 ms
len=28 ip=10.10.43.70 ttl=124 id=6039 icmp_seq=15 rtt=168.1 ms
len=28 ip=10.10.43.70 ttl=124 id=6040 icmp_seq=16 rtt=215.8 ms
```

A basic ICMP ping

2. hping3 -S -p 80 10.10.43.70

```
(kali㉿kali) - [~/Downloads]
$ sudo hping3 -S -p 80 10.10.43.70
HPING 10.10.43.70 (tun0 10.10.43.70): S set, 40 headers + 0 data bytes
len=40 ip=10.10.43.70 ttl=124 DF id=6047 sport=80 flags=RA seq=0 win=0 rtt=198.1 ms
len=40 ip=10.10.43.70 ttl=124 DF id=6048 sport=80 flags=RA seq=1 win=0 rtt=212.2 ms
len=40 ip=10.10.43.70 ttl=124 DF id=6049 sport=80 flags=RA seq=2 win=0 rtt=243.9 ms
len=40 ip=10.10.43.70 ttl=124 DF id=6050 sport=80 flags=RA seq=3 win=0 rtt=176.5 ms
len=40 ip=10.10.43.70 ttl=124 DF id=6051 sport=80 flags=RA seq=4 win=0 rtt=264.0 ms
len=40 ip=10.10.43.70 ttl=124 DF id=6052 sport=80 flags=RA seq=5 win=0 rtt=188.8 ms
len=40 ip=10.10.43.70 ttl=124 DF id=6053 sport=80 flags=RA seq=6 win=0 rtt=157.2 ms
len=40 ip=10.10.43.70 ttl=124 DF id=6054 sport=80 flags=RA seq=7 win=0 rtt=168.1 ms
len=40 ip=10.10.43.70 ttl=124 DF id=6055 sport=80 flags=RA seq=8 win=0 rtt=211.2 ms
^[[2~len=40 ip=10.10.43.70 ttl=124 DF id=6056 sport=80 flags=RA seq=10 win=0 rtt=204.2 ms
len=40 ip=10.10.43.70 ttl=124 DF id=6057 sport=80 flags=RA seq=11 win=0 rtt=161.4 ms
^[[len=40 ip=10.10.43.70 ttl=124 DF id=6058 sport=80 flags=RA seq=13 win=0 rtt=162.2 ms
```

Sends TCP SYN packets to port 80

3. hping3 -A -p 80 10.10.43.70

```
(kali㉿kali) - [~/Downloads]
$ sudo hping3 -A -p 80 10.10.43.70
HPING 10.10.43.70 (tun0 10.10.43.70): A set, 40 headers + 0 data bytes
len=40 ip=10.10.43.70 ttl=124 DF id=6145 sport=80 flags=R seq=0 win=0 rtt=203.1 ms
len=40 ip=10.10.43.70 ttl=124 DF id=6146 sport=80 flags=R seq=1 win=0 rtt=159.7 ms
len=40 ip=10.10.43.70 ttl=124 DF id=6147 sport=80 flags=R seq=2 win=0 rtt=185.5 ms
len=40 ip=10.10.43.70 ttl=124 DF id=6148 sport=80 flags=R seq=3 win=0 rtt=156.0 ms
len=40 ip=10.10.43.70 ttl=124 DF id=6149 sport=80 flags=R seq=4 win=0 rtt=191.6 ms
len=40 ip=10.10.43.70 ttl=124 DF id=6150 sport=80 flags=R seq=5 win=0 rtt=339.4 ms
len=40 ip=10.10.43.70 ttl=124 DF id=6151 sport=80 flags=R seq=6 win=0 rtt=176.7 ms
len=40 ip=10.10.43.70 ttl=124 DF id=6152 sport=80 flags=R seq=7 win=0 rtt=215.3 ms
len=40 ip=10.10.43.70 ttl=124 DF id=6153 sport=80 flags=R seq=8 win=0 rtt=158.6 ms
```

Sends TCP ACK packets to check if a host is alive

4. hping3 -2 -p 53 10.10.43.70

```
(kali㉿kali) - [~/Downloads]
$ sudo hping3 -2 -p 53 10.10.43.70
HPING 10.10.43.70 (tun0 10.10.43.70): udp mode set, 28 headers + 0 data bytes
ICMP Port Unreachable from ip=10.10.43.70 name=UNKNOWN
status=0 port=1338 seq=0
ICMP Port Unreachable from ip=10.10.43.70 name=UNKNOWN
status=0 port=1339 seq=1
ICMP Port Unreachable from ip=10.10.43.70 name=UNKNOWN
status=0 port=1340 seq=2
ICMP Port Unreachable from ip=10.10.43.70 name=UNKNOWN
status=0 port=1341 seq=3
ICMP Port Unreachable from ip=10.10.43.70 name=UNKNOWN
status=0 port=1342 seq=4
ICMP Port Unreachable from ip=10.10.43.70 name=UNKNOWN
status=0 port=1343 seq=5
ICMP Port Unreachable from ip=10.10.43.70 name=UNKNOWN
status=0 port=1344 seq=6
ICMP Port Unreachable from ip=10.10.43.70 name=UNKNOWN
status=0 port=1345 seq=7
ICMP Port Unreachable from ip=10.10.43.70 name=UNKNOWN
status=0 port=1346 seq=8
ICMP Port Unreachable from ip=10.10.43.70 name=UNKNOWN
status=0 port=1347 seq=9
54: ICMP Port Unreachable from ip=10.10.43.70 name=UNKNOWN
status=0 port=1348 seq=10
ICMP Port Unreachable from ip=10.10.43.70 name=UNKNOWN
status=0 port=1349 seq=11
AC
--- 10.10.43.70 hping statistic ---
13 packets transmitted, 12 packets received, 8% packet loss
round-trip min/avg/max = 154.2/193.5/448.6 ms
```

Uses UDP packets to a specified port

5. hping3 --flood -1 10.10.43.70

```
(kali㉿kali) - [~/Downloads]
$ sudo hping3 --flood -1 10.10.43.70
HPING 10.10.43.70 (tun0 10.10.43.70): icmp mode set, 28 headers + 0 data bytes
hping in flood mode, no replies will be shown
```

Flooding the network by sending ICMP request rapidly.